Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 3 Tutorial Worksheet: Layer 2 Attacks

Field	Detail
Based on	Week 3, Layer 2 Attacks lecture (VLANs, CAM table, MAC address spoofing, VLAN hopping, DHCP, and ARP attacks)
Learning Outcomes	LO1 (describe network vulnerabilities/attacks) and LO2 (evaluate mitigation techniques)
Format	14 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's lecture. Several ask you to evaluate a mitigation, not just describe an attack. That's deliberate: LO2 is specifically about weighing mitigation options, not just naming them. Write full answers, not one-line definitions.

VLAN Foundations

Q1. Explain what a VLAN is and why segmenting a network into VLANs is a security benefit, not just an organisational convenience. *[LO1]*

Q2. Explain what the native VLAN is, and why leaving it as the default (VLAN 1) is considered a security risk. *[LO1]*

CAM Table Attack

Q3. Explain how a CAM table (MAC flooding) attack works, and why a switch's 'fail open' design is what makes it possible. *[LO1]*

Q4. A network administrator is deciding between two options: (a) leave port security off everywhere, or (b) enable port security with a maximum of 2 MAC addresses and violation mode 'restrict' on every access port. Evaluate this trade-off. What does option (b) cost operationally, and is it worth it? *[LO2]*

MAC Address Spoofing

Q5. Explain how MAC address spoofing differs from a CAM table attack, even though both exploit the same underlying trust in MAC addresses as a reliable identity. *[LO1]*

Q6. A company relies on MAC-based port security (sticky MAC, no other authentication) as its only defence against unauthorised devices. Evaluate why this alone is insufficient against MAC address spoofing specifically, and recommend a stronger alternative. *[LO2]*

VLAN Hopping

Q7. Describe how switch spoofing works, and explain why disabling DTP (switchport nonegotiate) is such an effective, low-cost fix. *[LO1, LO2]*

Q8. Describe how the VLAN double-tagging attack works. Why does this attack only work in one direction (attacker to target VLAN), and why does moving the native VLAN off VLAN 1 mitigate it? *[LO1, LO2]*

DHCP-Based Attacks

Q9. Distinguish between DHCP starvation and DHCP spoofing. What does each attack actually achieve for the attacker, and how do the techniques differ? *[LO1]*

Q10. Explain how DHCP Snooping mitigates both DHCP starvation and DHCP spoofing, and why the trusted/untrusted port distinction is central to how it works. *[LO2]*

ARP Spoofing

Q11. Explain how ARP spoofing sets up a man-in-the-middle position between two victims, and why hosts accepting unsolicited ARP replies is the root cause. *[LO1]*

Q12. Explain why Dynamic ARP Inspection has to be configured after DHCP Snooping, not before or independently. What would happen if you enabled DAI without DHCP Snooping already running? *[LO2]*

Bringing It Together

Q13. Several attacks covered this week can produce more than one outcome depending on attacker intent. For example, a CAM table attack can serve as either quiet eavesdropping or a performance-degrading denial of service. Choose two other attacks from this week, state both plausible outcomes for each, and explain why the same technique can be aimed at different goals. *[LO1, LO2]*

Q14. A small company's network has no Layer 2 security controls configured at all. Using this week's seven attacks, prioritise the top three you'd fix first and justify your ordering. Consider both risk and ease of mitigation. *[LO2, LO3]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. This week's lab is direct hands-on practice for several of these mitigations. Configuring them yourself will cement the theory faster than rereading it.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5). The comparison/evaluation-style questions above are close in format to the structured-essay section.